Name
Date
Course

Converting between the Three Forms of a Quadratic Expression

The Three Forms of a Quadratic Expression

Standard Form	Intercept Form	Vertex Form
$y = ax^2+bx+c$	$y=a(x-m)(x-n)$	$y=a(x-h)^2+k$

Write the form of each quadratic expression below and convert to the other two forms.

① $y=-2(x-1)^2+8$

I) $\boxed{\text{Vertex}}$

II) Standard

$y=-2(x-1)^2+8$

$y=-2(x-1)(x-1)+8$

$y=-2(x^2-x-x+1)+8$

$y=-2(x^2-2x+1)+8$

$y=-2x^2+4x-2+8$

$\boxed{y=-2x^2+4x+6}$

III) Intercept

$y=-2x^2+4x+6$

$y=-2(x^2-2x-3)$

$y=\boxed{-2(x-3)(x+1)}$

② $6(x+3)^2-24=y$

I) $\boxed{\text{Vertex}}$

II) Standard

$6(x+3)^2-24=y$

$6(x+3)(x+3)-24=y$

$6(x^2+3x+3x+9)-24=y$

$6(x^2+6x+9)-24=y$

$6x^2+36x+54-24=y$

$\boxed{6x^2+36x+30=y}$

III) Intercept

$6x^2+36x+30=y$

$6(x^2+6x+5)=y$

$\boxed{6(x+5)(x+1)=y}$

③ $(x-4)^2-4=y$

I) $\boxed{\text{Vertex}}$

II) Standard

$(x-4)^2-4=y$

$(x-4)(x-4)-4=y$

$x^2-4x-4x+16-4=y$

$\boxed{x^2-8x+12=y}$

III) Intercept

$x^2-8x+12=y$

$\boxed{(x-2)(x-6)=y}$

④ $y=2(x-1)^2-18$

I) $\boxed{\text{Vertex}}$

II) Standard

$y=2(x-1)^2-18$

$y=2(x-1)(x-1)-18$

$y=2(x^2-x-x+1)-18$

$y=2(x^2-2x+1)-18$

$y=2x^2-4x+2-18$

$\boxed{y=2x^2-4x-16}$

III) Intercept

$y=2x^2-4x-16$

$y=2(x^2-2x-8)$

$\boxed{y=2(x-4)(x+2)}$

⑤ $-3(x-6)^2+3=y$

I) $\boxed{\text{Vertex}}$

II) Standard

$-3(x-6)^2+3=y$

$-3(x-6)(x-6)+3=y$

$-3(x^2-6x-6x+36)+3=y$

$-3(x^2-12x+36)+3=y$

$-3x^2+36x-108+3=y$

$\boxed{-3x^2+36x-105=y}$

III) Intercept

$-3x^2+36x-105=y$

$-3(x^2-12x+35)=y$

$\boxed{-3(x-7)(x-5)=y}$

⑥ $(x+1)^2-25=y$

I) $\boxed{\text{Vertex}}$

II) Standard

$(x+1)^2-25=y$

$(x+1)(x+1)-25=y$

$x^2+x+x+1-25=y$

$\boxed{x^2+2x-24=y}$

III) Intercept

$x^2+2x-24=y$

$\boxed{(x+6)(x-4)=y}$

⑦ $y = 5(x-5)(x-1)$

I) $\boxed{\text{Intercept}}$

II) Standard

$y = 5(x-5)(x-1)$

$y = 5(x^2 - x - 5x + 5)$

$y = 5(x^2 - 6x + 5)$

$\boxed{y = 5x^2 - 30x + 25}$

III) Vertex

$y = 5x^2 - 30x + 25$

$\frac{y}{5} = \frac{5x^2 - 30x + 25}{5}$

$\frac{y}{5} = x^2 - 6x + 5$

$\frac{y}{5} - 5 = x^2 - 6x$

$\frac{y}{5} - 5 + 9 = x^2 - 6x + 9$

$\frac{y}{5} + 4 = (x-3)(x-3)$

$\frac{y}{5} + 4 = (x-3)^2$

$\frac{y}{5} = (x-3)^2 - 4$

$(5)\frac{y}{5} = (5)((x-3)^2 - 4)$

$\boxed{y = 5(x-3)^2 - 20}$

⑧ $-2(x+3)(x+1) = y$

I) $\boxed{\text{Intercept}}$

II) Standard

$-2(x+3)(x+1) = y$

$-2(x^2 + x + 3x + 3) = y$

$-2(x^2 + 4x + 3) = y$

$\boxed{-2x^2 - 8x - 6 = y}$

III) Vertex

$-2x^2 - 8x - 6 = y$

$\frac{-2x^2 - 8x - 6}{-2} = \frac{y}{-2}$

$x^2 + 4x + 3 = \frac{y}{-2}$

$x^2 + 4x = \frac{y}{-2} - 3$

$x^2 + 4x + 4 = \frac{y}{-2} - 3 + 4$

$(x+2)(x+2) = \frac{y}{-2} + 1$

$(x+2)^2 = \frac{y}{-2} + 1$

$(x+2)^2 - 1 = \frac{y}{-2}$

$(-2)((x+2)^2 - 1) = \frac{y}{-2}(-2)$

$\boxed{-2(x+2)^2 + 2 = y}$

⑨ $y=(x+1)(x-9)$

I) $\boxed{\text{Intercept}}$

II) Standard

$y=(x+1)(x-9)$

$y=x^2-9x+x-9$

$\boxed{y=x^2-8x-9}$

III) Vertex

$y=x^2-8x-9$

$y+9=x^2-8x$

$y+9+16=x^2-8x+16$

$y+25=(x-4)(x-4)$

$y+25=(x-4)^2$

$\boxed{y=(x-4)^2-25}$

⑩ $-4(x+2)(x-4)=y$

I) $\boxed{\text{Intercept}}$

II) Standard

$-4(x+2)(x-4)=y$

$-4(x^2-4x+2x-8)=y$

$-4(x^2-2x-8)=y$

$\boxed{-4x^2+8x+32=y}$

III) Vertex

$-4x^2+8x+32=y$

$\dfrac{-4x^2+8x+32}{-4}=\dfrac{y}{-4}$

$x^2-2x-8=\dfrac{y}{-4}$

$x^2-2x+1=\dfrac{y}{-4}+8+1$

$(x-1)(x-1)=\dfrac{y}{-4}+9$

$(x-1)^2=\dfrac{y}{-4}+9$

$(x-1)^2-9=\dfrac{y}{-4}$

$(-4)((x-1)^2-9)=\dfrac{y}{-4}(-4)$

$\boxed{-4(x-1)^2+36=y}$

(11) $y = 3(x+2)(x+6)$

I) $\boxed{\text{Intercept}}$

II) Standard

$y = 3(x+2)(x+6)$

$y = 3(x^2+6x+2x+12)$

$y = 3(x^2+8x+12)$

$\boxed{y = 3x^2+24x+36}$

III) Vertex

$y = 3x^2+24x+36$

$\frac{y}{3} = \frac{3x^2+24x+36}{3}$

$\frac{y}{3} = x^2+8x+12$

$\frac{y}{3} - 12 = x^2+8x$

$\frac{y}{3} - 12 + 16 = x^2+8x+16$

$\frac{y}{3} + 4 = (x+4)(x+4)$

$\frac{y}{3} + 4 = (x+4)^2$

$\frac{y}{3} = (x+4)^2 - 4$

$(3)\frac{y}{3} = (3)\left((x+4)^2 - 4\right)$

$\boxed{y = 3(x+4)^2 - 12}$

(12) $-(x-3)(x-7) = y$

I) $\boxed{\text{Intercept}}$

$\boxed{-(x-5)^2+4 = y}$

II) Standard

$-(x-3)(x-7) = y$

$-(x^2-7x-3x+21) = y$

$-(x^2-10x+21) = y$

$\boxed{-x^2+10x-21 = y}$

III) Vertex

$-x^2+10x-21 = y$

$\frac{-x^2+10x-21}{-1} = \frac{y}{-1}$

$x^2-10x+21 = -y$

$x^2-10x+25 = -y-21+25$

$(x-5)(x-5) = -y+4$

$(x-5)^2 = -y+4$

$(x-5)^2 - 4 = -y$

$(-1)\left((x-5)^2-4\right) = -y(-1)$

⑬ $2x^2-12x+16=y$

I) $\boxed{\text{Standard}}$

II) Intercept

$2x^2-12x+16=y$

$2(x^2-6x+8)=y$

$\boxed{2(x-4)(x-2)=y}$

III) Vertex

$2x^2-12x+16=y$

$\frac{2x^2-12x+16}{2}=\frac{y}{2}$

$x^2-6x+8=\frac{y}{2}$

$x^2-6x=\frac{y}{2}-8$

$x^2-6x+9=\frac{y}{2}-8+9$

$(x-3)(x-3)=\frac{y}{2}+1$

$(x-3)^2=\frac{y}{2}+1$

$(x-3)^2-1=\frac{y}{2}$

$(x-3)^2-1=\frac{y}{2}$

$(2)((x-3)^2-1)=\frac{y}{2}(2)$

$\boxed{2(x-3)^2-2=y}$

⑭ $y=-5x^2+15x+50$

I) $\boxed{\text{Standard}}$

II) Intercept

$y=-5x^2+15x+50$

$y=-5(x^2-3x-10)$

$\boxed{y=-5(x-5)(x+2)}$

III) Vertex

$y=-5x^2+15x+50$

$\frac{y}{-5}=\frac{-5x^2+15x+50}{-5}$

$\frac{y}{-5}=x^2-3x-10$

$\frac{y}{-5}+10=x^2-3x$

$\frac{y}{-5}+10+\frac{9}{4}=x^2-3x+\frac{9}{4}$

$\frac{y}{-5}+\frac{49}{4}=(x-\frac{3}{2})(x-\frac{3}{2})$

$\frac{y}{-5}+\frac{49}{4}=(x-\frac{3}{2})^2$

$\frac{y}{-5}=(x-\frac{3}{2})^2-\frac{49}{4}$

$(-5)\frac{y}{-5}=(-5)((x-\frac{3}{2})^2-\frac{49}{4})$

$\boxed{y=-5(x-\frac{3}{2})^2+\frac{245}{4}}$

(15) $x^2+10x+9=y$

I) $\boxed{\text{Standard}}$

II) Intercept

$x^2+10x+9=y$

$\boxed{(x+9)(x+1)=y}$

III) Vertex

$x^2+10x+9=y$

$x^2+10x+25=y-9+25$

$(x+5)(x+5)=y+16$

$(x+5)^2=y+16$

$\boxed{(x+5)^2-16=y}$

(16) $y=-3x^2-18x-15$

I) $\boxed{\text{Standard}}$

II) Intercept

$y=-3x^2-18x-15$

$y=-3(x^2+6x+5)$

$\boxed{y=-3(x+5)(x+1)}$

III) Vertex

$y=-3x^2-18x-15$

$\frac{y}{-3}=\frac{-3x^2-18x-15}{-3}$

$\frac{y}{-3}=x^2+6x+5$

$\frac{y}{-3}-5=x^2+6x$

$\frac{y}{-3}-5+9=x^2+6x+9$

$\frac{y}{-3}+4=(x+3)(x+3)$

$\frac{y}{-3}+4=(x+3)^2$

$\frac{y}{-3}=(x+3)^2-4$

$(-3)\frac{y}{-3}=-3((x+3)^2-4)$

$\boxed{y=-3(x+3)^2+12}$

⑰ $4x^2-32x+60=y$

I) $\boxed{\text{Standard}}$

II) Intercept

$$4x^2-32x+60=y$$

$$4(x^2-8x+15)=y$$

$$\boxed{4(x-5)(x-3)=y}$$

III) Vertex

$$4x^2-32x+60=y$$

$$\frac{4x^2-32x+60}{4}=\frac{y}{4}$$

$$x^2-8x+15=\frac{y}{4}$$

$$x^2-8x=\frac{y}{4}-15$$

$$x^2-8x+16=\frac{y}{4}-15+16$$

$$(x-4)(x-4)=\frac{y}{4}+1$$

$$(x-4)^2=\frac{y}{4}+1$$

$$(x-4)^2=\frac{y}{4}+1$$

$$(x-4)^2-1=\frac{y}{4}$$

$$(4)\left((x-4)^2-1\right)=\frac{y}{4}(4)$$

$$\boxed{4(x-4)^2-4=y}$$

⑱ $y=x^2+9x+14$

I) $\boxed{\text{Standard}}$

II) Intercept

$$y=x^2+9x+14$$

$$\boxed{y=(x+7)(x+2)}$$

III) Vertex

$$y=x^2+9x+14$$

$$y-14=x^2+9x$$

$$y-14+\frac{81}{4}=x^2+9x+\frac{81}{4}$$

$$y+\frac{25}{4}=\left(x+\frac{9}{2}\right)\left(x+\frac{9}{2}\right)$$

$$y+\frac{25}{4}=\left(x+\frac{9}{2}\right)^2$$

$$\boxed{y=\left(x+\frac{9}{2}\right)^2-\frac{25}{4}}$$